

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS:

Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. <div>
- B. <nav>
- C. <section>
- D. <header>

Snippet 4: XML Syntax

- A. <note><to>User</to><from>Admin</from>
- B. <note><to>User</to><from>Admin</from></note>
- C. <note><to>User</from></note>
- D. <note to="User"></note>

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I N.Loka Bandhava (192324167)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

N.Loka Bandhava
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CSA4305-Internet Programming

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Scenario: Web Development Standards Implementation

A startup building a modern web application encountered issues spanning HTML elements, HTTP methods, XML syntax, and web communication protocols. Each question below is answered against the provided data snippets, with the correct option identified and justified, and the incorrect options explained.

Question 1: Video Embedding

Snippet 1 options:

A. `<media src="video.mp4"></media>`[Text Wrapping Break] B. `<video src="video.mp4" controls></video>`[Text Wrapping Break] C. `<movie src="video.mp4"></movie>`[Text Wrapping Break] D. `<vid src="video.mp4"></vid>`

Correct Answer: B — `<video src="video.mp4" controls></video>`

Justification

- ✗ `<video>` is the official HTML5 semantic element specifically defined by the W3C/WHATWG standard for embedding video content in a web page.
- ✗ It is natively understood and rendered by all modern browsers without needing external plugins (unlike the old `<embed></object>` era).
- ✗ The controls attribute automatically provides the browser's built-in play/pause, volume, seek bar, and fullscreen UI, improving usability with no extra JavaScript.
- ✗ It supports accessible fallback content, multiple `<source>` children for format negotiation, and attributes like autoplay, loop, muted, and poster.

Why the other options are incorrect

- ✗ A. `<media>` — not a valid or recognized HTML tag; browsers do not know how to render it, so it falls back to an inline, unstyled unknown element with no video functionality.
- ✗ C. `<movie>` — not part of the HTML specification at all; it is not recognized by any browser rendering engine.
- ✗ D. `<vid>` — likewise not a standard HTML element; using a shortened, non-standard tag name provides no built-in video decoding, controls, or accessibility support.

Question 2: Form Submission Method

Snippet 2 options:

A. GET[Text Wrapping Break] B. POST[Text Wrapping Break] C. PUT[Text Wrapping Break] D. DELETE

Correct Answer: B — POST

Justification

- ✗ The registration form must send user data (name, email, password, etc.) securely to the server, so POST is appropriate because it transmits data in the HTTP request body rather than the URL.
- ✗ Data sent via POST does not appear in the browser address bar, browser history, or server access logs the way GET query strings do.
- ✗ POST has no meaningful size restriction on the payload, so it comfortably handles registration forms with many fields or file uploads.
- ✗ Semantically, submitting a registration form creates a new resource (a new user account) on the server — this matches POST's defined purpose of submitting data to be processed, as opposed to GET's purpose of safely retrieving a resource.

Why the other options are incorrect

- ⊘ A. GET — would expose all submitted form fields, including sensitive data, directly in the URL; unsuitable and insecure for registration/login data.
- ⊘ C. PUT — is used to create or completely replace an existing, identified resource at a known URL (idempotent updates via an API), not to submit a new HTML form from a browser; standard HTML forms do not even support method="PUT" natively.
- ⊘ D. DELETE — is used to remove a resource from the server; it has no role in submitting new registration data.

Question 3: Navigation Element

Snippet 3 options:

A. `<div>`[Text Wrapping Break]B. `<nav>`[Text Wrapping Break]C. `<section>`[Text Wrapping Break]D. `<header>`

Correct Answer: B — `<nav>`

Justification

- ⊘ `<nav>` is the HTML5 semantic element specifically designed to wrap a block of primary navigation links (main menu, table of contents, breadcrumbs, etc.).
- ⊘ It clearly communicates the purpose of its content to browsers, search engine crawlers, and assistive technologies such as screen readers, which can let users jump directly to the navigation region.
- ⊘ It improves SEO, since search engines can distinguish site navigation from the actual page content when indexing.
- ⊘ It keeps the document outline meaningful and self-describing compared to a generic container.

Why the other options are incorrect

- ⊘ A. `<div>` — a generic, non-semantic container; it groups elements visually/structurally but conveys no meaning about what the content represents, forcing assistive tools to rely on extra ARIA roles or class-name guessing.
- ⊘ C. `<section>` — represents a standalone thematic grouping of content (usually with its own heading), such as an article section — it is not intended specifically for navigation links.
- ⊘ D. `<header>` — represents introductory content or a group of introductory/navigational aids for its nearest ancestor, typically containing a logo, title, and sometimes a `<nav>`, but it is not itself the semantic element for the menu — the menu links still belong inside a `<nav>`, ideally placed within the `<header>`.

Question 4: XML Syntax

Snippet 4 options:

A. `<note><to>User</to><from>Admin</from>`[Text Wrapping Break]B. `<note><to>User</to><from>Admin</from></note>`[Text Wrapping Break]C. `<note><to>User</from></note>`[Text Wrapping Break]D. `<note to="User"></note>`

Correct Answer: B — `<note><to>User</to><from>Admin</from></note>`

Justification

XML requires every document to be well-formed: it must have exactly one root element, and every opening tag must have a matching closing tag with correctly nested, non-overlapping elements. Option B satisfies all of these rules — `<note>` is the single root element, and both of its children, `<to>` and `<from>`, are properly opened and closed in the correct order before the root itself is closed.

Why the other options are incorrect

- ⊘ A. `<note><to>User</to><from>Admin</from>` — the root element `<note>` is never closed; the document is missing its final `</note>` tag, so it is not well-formed.

- ⊄ C. `<note><to>User</from></note>` — mismatched tags: the element is opened as `<to>` but closed as `</from>`, violating XML's rule that a closing tag must exactly match the most recently opened tag (improper nesting).
- ⊄ D. `<note to="User"></note>` — the closing tag is missing its final `>`, so the markup itself is syntactically broken/incomplete and will not parse.

Question 5: Web Communication Protocol

Snippet 5 options:

A. FTP[Text Wrapping Break]B. HTTP[Text Wrapping Break]C. SMTP[Text Wrapping Break]D. SNMP

Correct Answer: B — HTTP (or HTTPS)

Justification

- ⊄ HTTP (HyperText Transfer Protocol) — and its encrypted form, HTTPS — is the application-layer protocol specifically designed for transferring web pages, forms, and other resources between browsers (clients) and web servers.
- ⊄ It defines the request/response model (GET, POST, PUT, DELETE, etc.), status codes, and headers that underlie all standard web application communication, exactly matching what this startup's application needs.
- ⊄ For any production application handling registration and user data, HTTPS specifically should be used, as it layers TLS encryption on top of HTTP to protect data in transit.

Why the other options are incorrect

- ⊄ A. FTP — File Transfer Protocol is designed for uploading/downloading files between a client and a file server; it is not used for serving interactive web pages or handling HTTP-based form submissions.
- ⊄ C. SMTP — Simple Mail Transfer Protocol is used specifically for sending email between mail servers; it has no role in browser-to-web-server communication.
- ⊄ D. SNMP — Simple Network Management Protocol is used for monitoring and managing network devices (routers, switches); it is unrelated to serving or interacting with a web application.